Stochastic Gradient Descent

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August 8, 2025

Outline

1. Motivation for SGD
2. Stochastic Gradient Descent
3. Examples

Motivation

Deterministic algorithms:

$$x \longrightarrow \boxed{\text{oracle}} \longrightarrow \nabla f(x)$$

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Stochastic gradient descent (SGD)

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In large scale machine learning, using **SGD** is often a better choice due to computational efficiency.

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  $$a_i \in \mathbb{R}^d, \quad b_i \in \{\pm 1\}$$

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  $$f(x) = \frac{1}{n}\sum_{i=1}^{n}\log\left(1 + e^{b_i a_i^\top x}\right) \qquad \text{[Binary classification]}$$

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With this, define:

$$\mathcal{G}(x; \xi) = \nabla f_\xi(x) \qquad \text{(Random variable because of } \xi\text{)}$$

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- **Quality of** $\mathcal{G}(x,\xi)$ **to be controlled.**(variance control)

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- **Boundedness Assumption:** We assume a structure on the second moment:

$$\mathbb{E}_\xi\left[\|G(x;\xi)\|^2\right] \le B^2$$

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  - **Boundedness assumption:**Assume f is $L-$Lipscitz, then describe the boundedness assumption(Exercise!)

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where $e_\xi^\top = [0 \ \dots \ 1 \ \dots \ 0]$ is the standard basis vector.

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Remark: This is useful, since we expect $\|x - x^*\|$ to be small ,implying $\mathbb{E}_\xi\left[\|G(x;\xi)\|\right]$ is small.

Thank You!